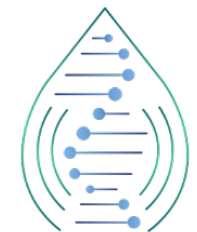


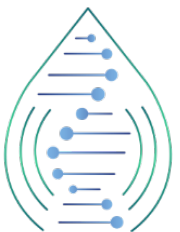
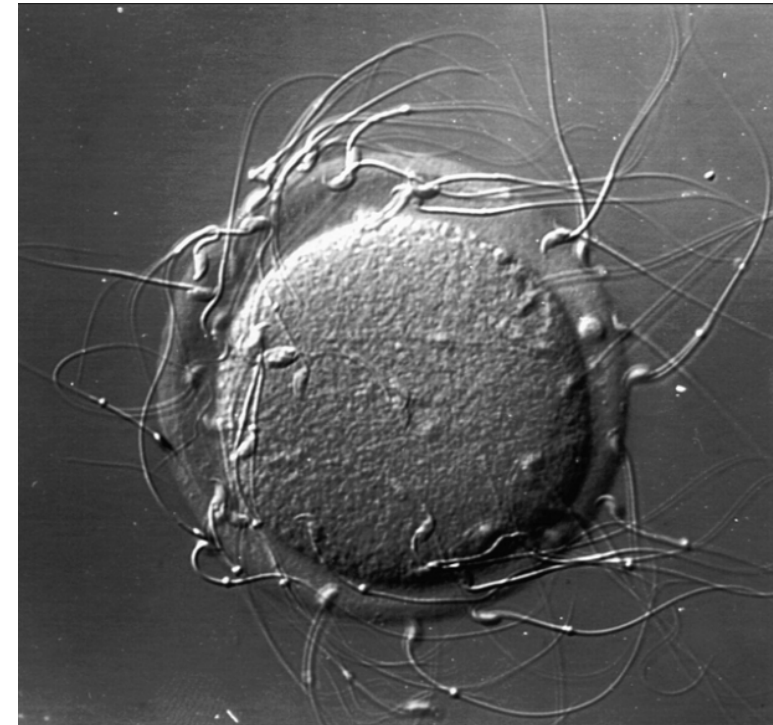
CRISPR-Cas 9 genome editing in human embryos – basic research and potential clinical application.

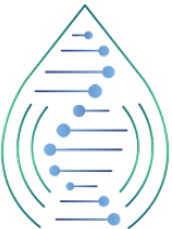
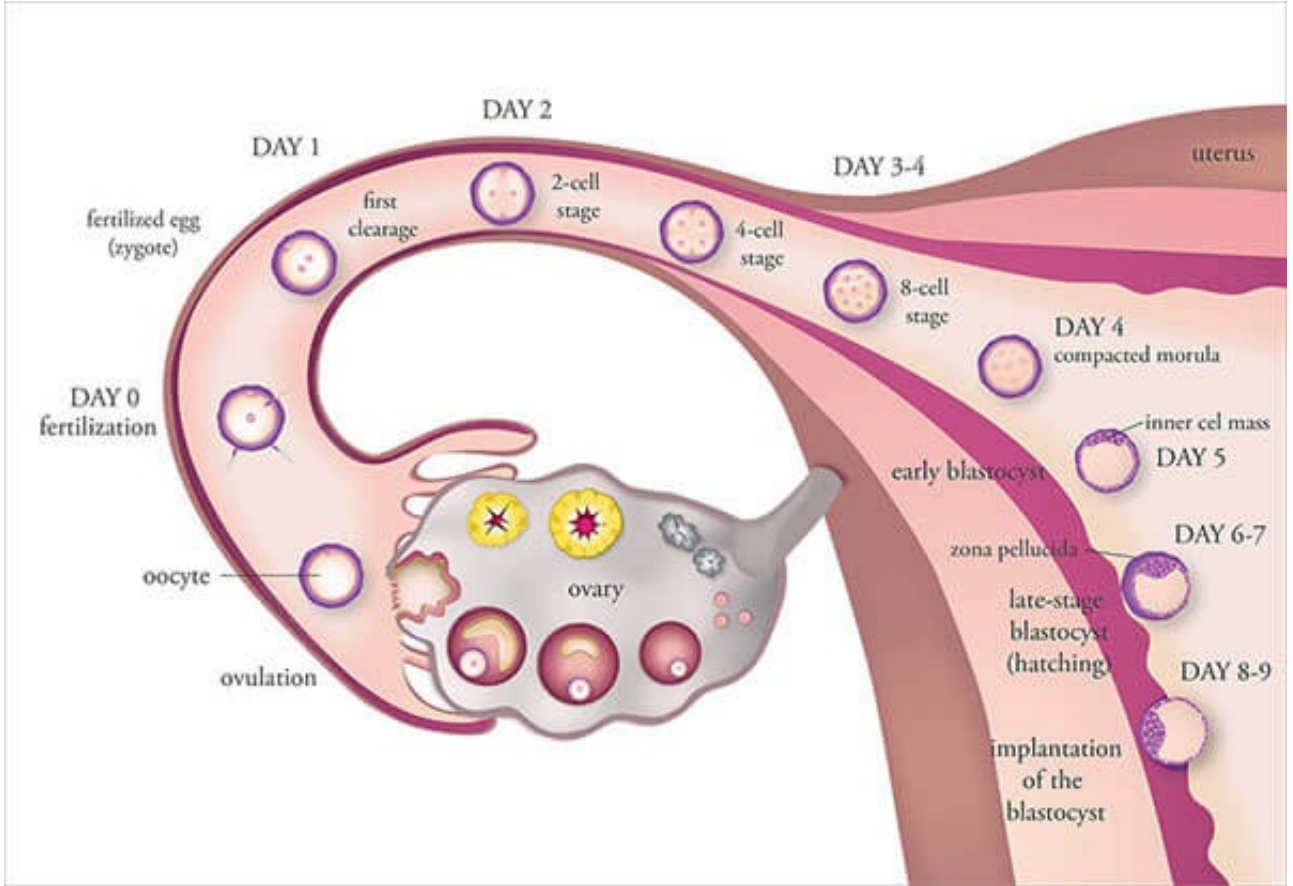
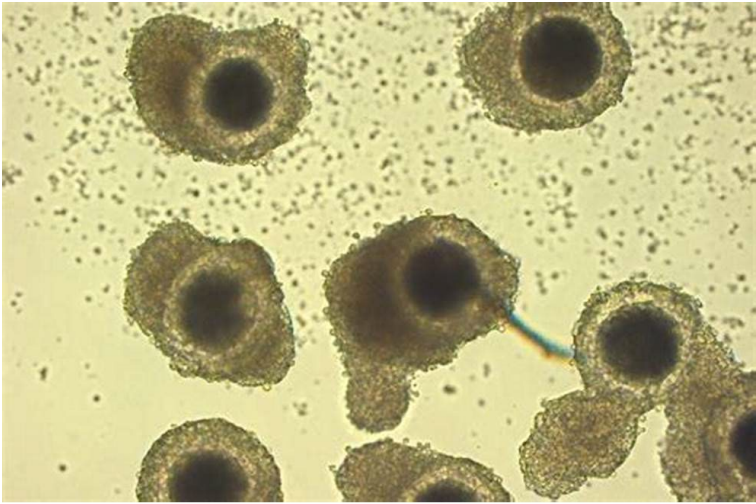
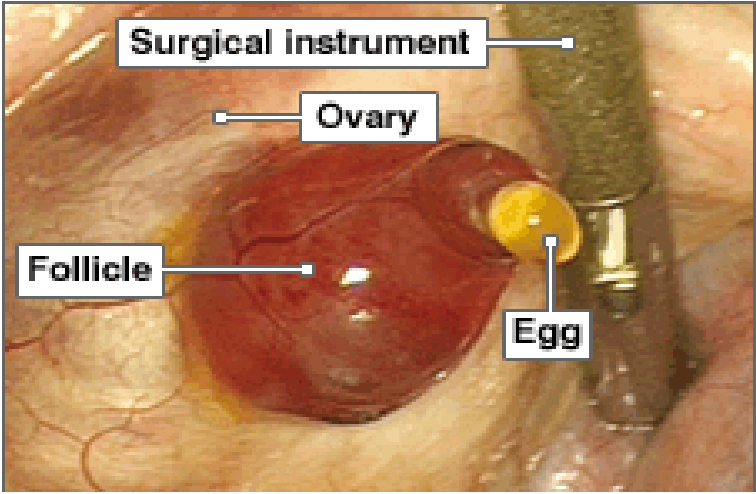
Munuse C Savash

*Oxford University
Life Research Oxford*

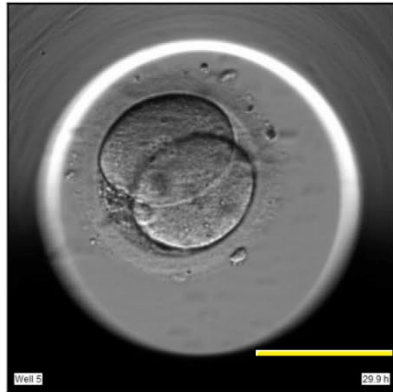


Introduction to embryology

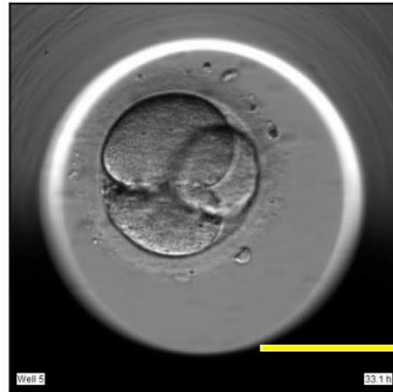




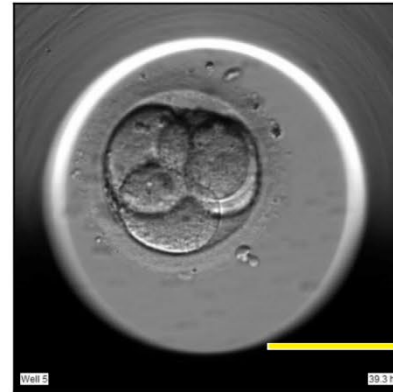
The human embryo



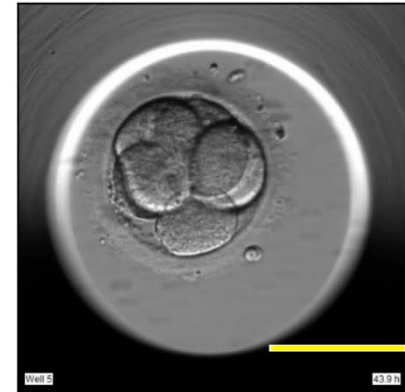
2-cell



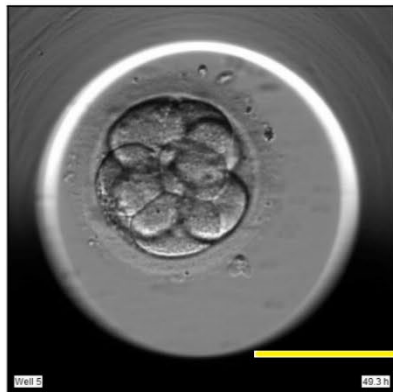
3-cell



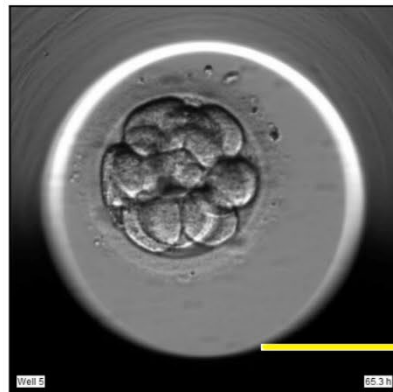
4-cell



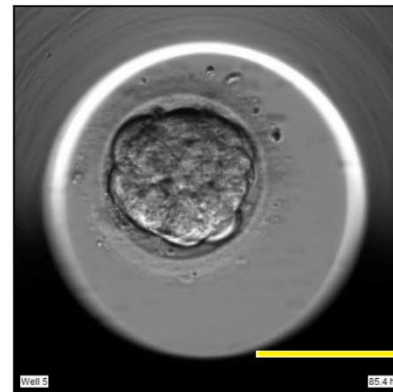
5-cell



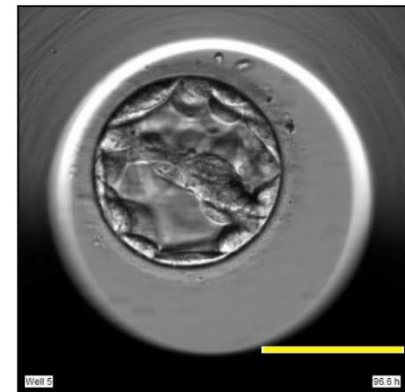
8-cell



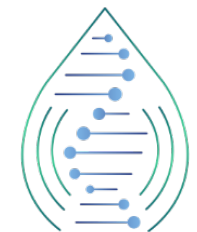
9+-cell



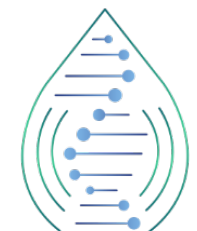
morula

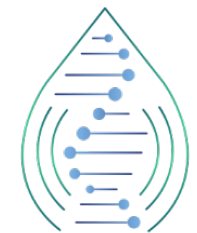
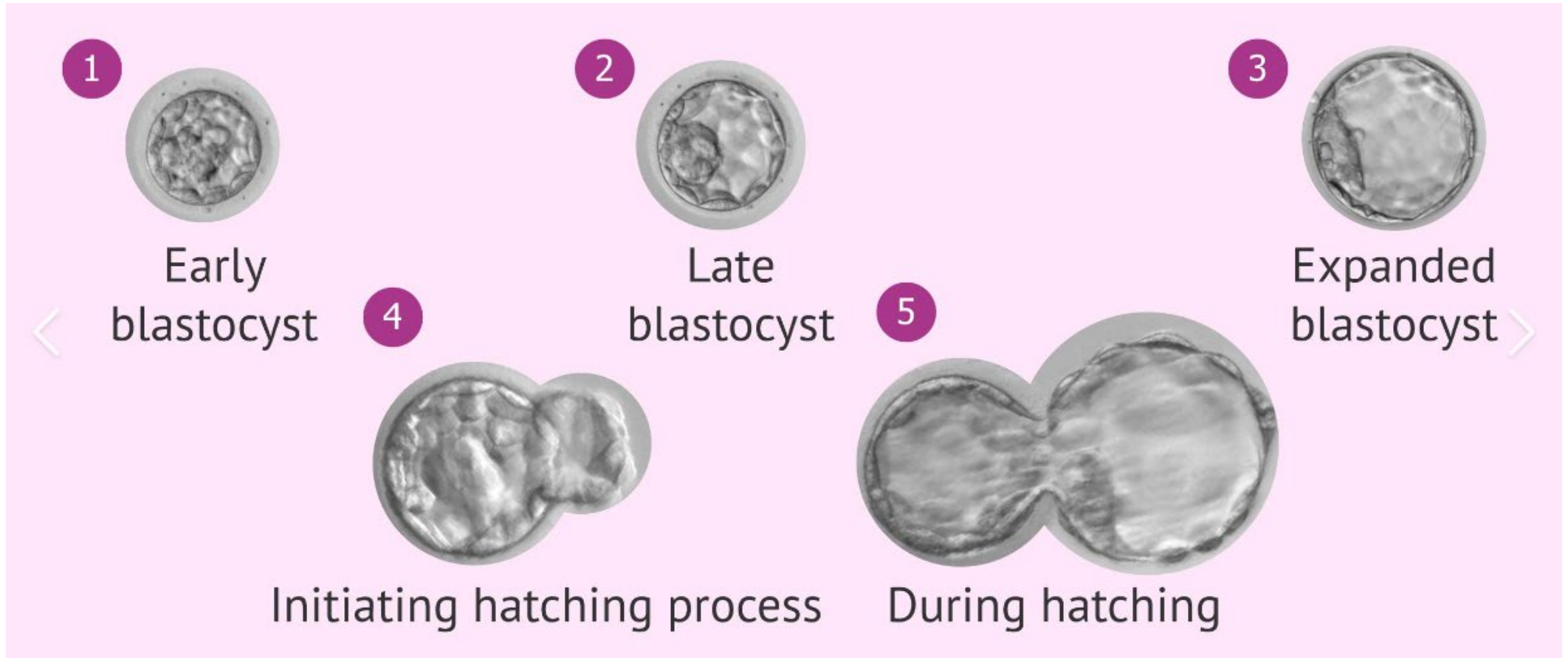


blastocyst



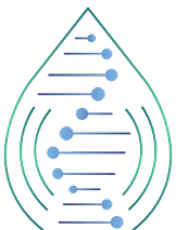
Blastocyst





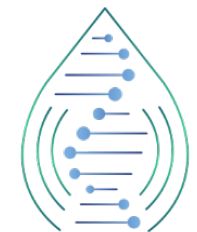
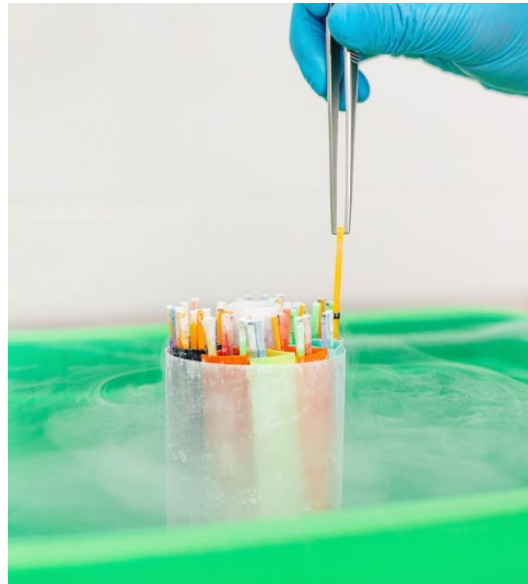
IVF

- Developed by Dr Robert Edwards, Dr Patrick Steptoe and Jean Purdy.
- The first IVF baby was born in 1978.
- Dr Edwards was awarded the Nobel prize in Medicine in 2010.

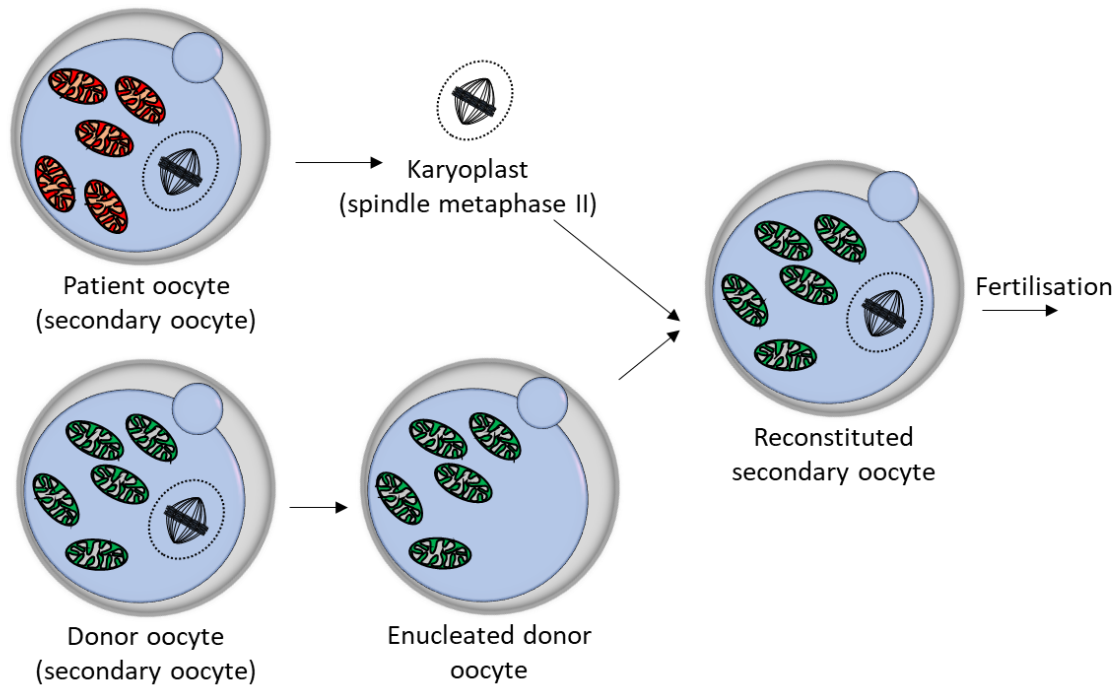


Advancements of IVF

- Genetic screening
- Non-invasive genetic screening
- Intracytoplasmic sperm injection
- Cryopreservation
- IVM
- MST ?
- Gene editing ?

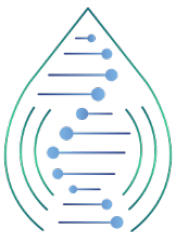


MST

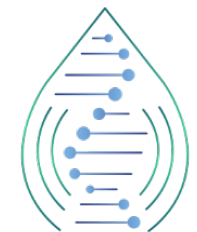


First pilot study of maternal spindle transfer for the treatment of repeated in vitro fertilization failures in couples with idiopathic infertility

Nuno Costa-Borges, Ph.D.,^{a,1} Eros Nikitos, M.Sc.,^{b,1} Katharina Späth, Ph.D.,^{c,1} Irene Miguel-Escalada, Ph.D.,^{a,*} Hong Ma, Ph.D.,^d Klaus Rink, Ph.D.,^a Clement Coudereau, Ph.D.,^c Hayley Darby,^d Amy Koski, M.Sc.,^d Crystal Van Dyken, Ph.D.,^d Enric Mestres, Ph.D.,^a Evmorfia Papakyriakou, M.Sc.,^b Dominique De Ziegler, M.D.,^b George Kontopoulos, M.D.,^b Themistoklis Mantzavinos, M.D.,^b Ioannis Vasilopoulos, M.D.,^b Stylianos Grigorakis, M.D.,^b Thomas Prokopakis, M.D.,^b Konstantinos Dimitropoulos, M.D.,^b Panagiotis Polyzos, M.D.,^b Nikolas Vlachos, M.D.,^b Konstantinos Kostaras, M.D.,^b Shoukhrat Mitalipov, Ph.D.,^d Gloria Calderón, Ph.D.,^a Panagiotis Psathas, M.D.,^b and Dagan Wells, Ph.D.,^{c,e}



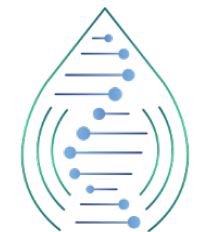
Germline gene editing



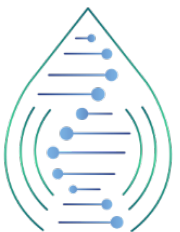
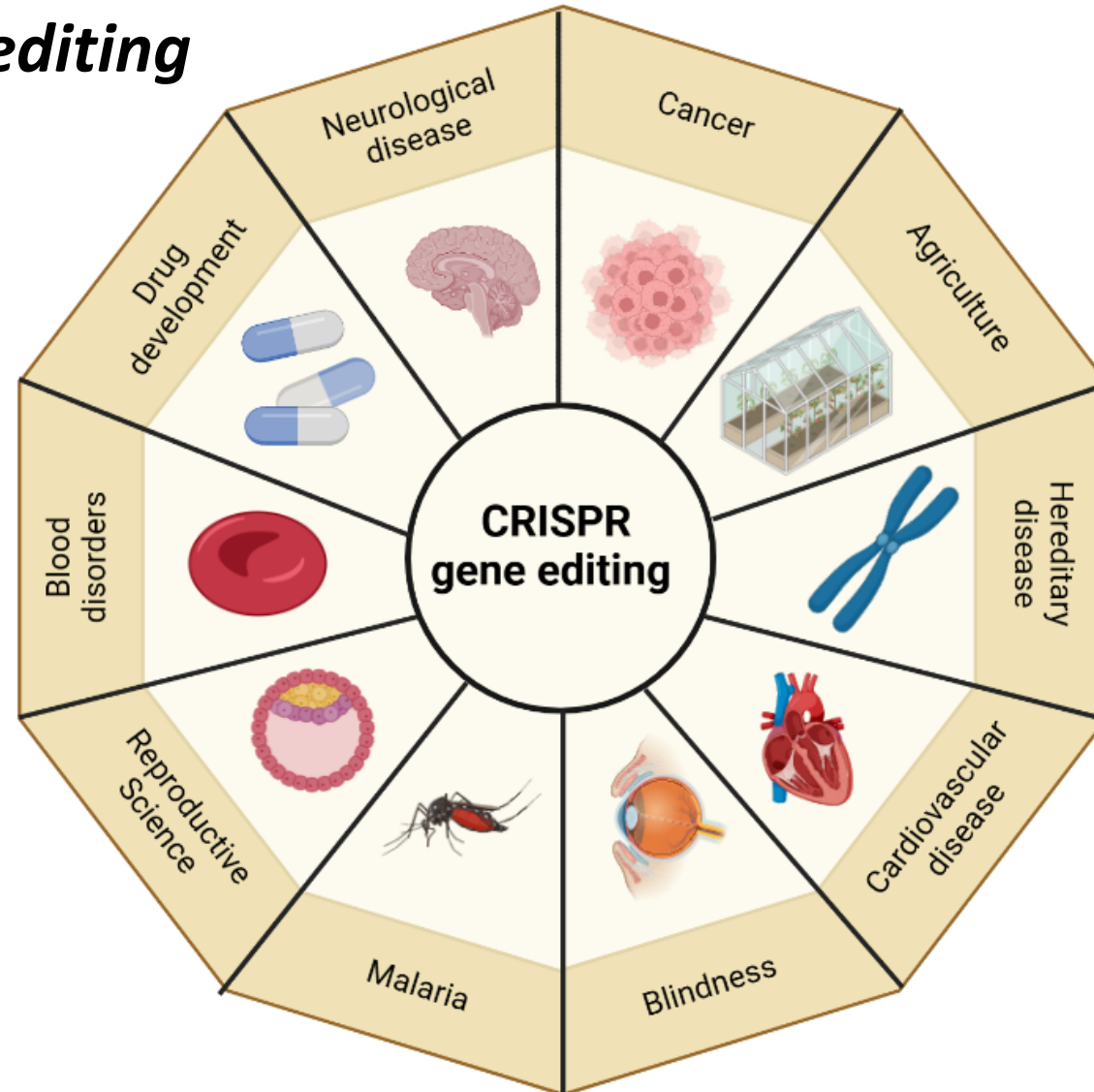
Founders



Founded by Dr Jennifer Doudna and Dr Emmanuelle Charpentier.
Nobel prize winners in 2020.



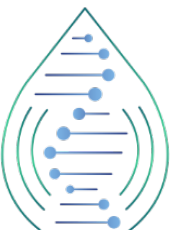
Current uses of gene editing



Why edit embryos?

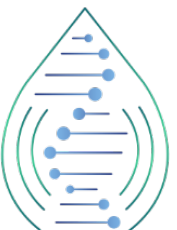
What are some of the reasons we may want to edit embryos?

1. Understand embryonic development.
2. Studying causes of infertility and miscarriage.
3. Investigate genetic disorders



What are the regulations behind editing embryos?

- Require a special licence from the HFEA
- Cannot culture the embryos for more than 14 days
- Research purposes only!!!
- Only using **spare** embryos donated from IVF



How is CRISPR introduced into an embryo?

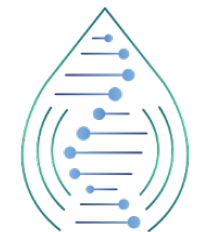
What are some current methods of inserting the CRISPR complex into cells ?

Chemical transfection

Electroporation

Lipofection

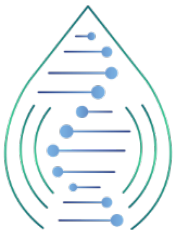
Microinjection

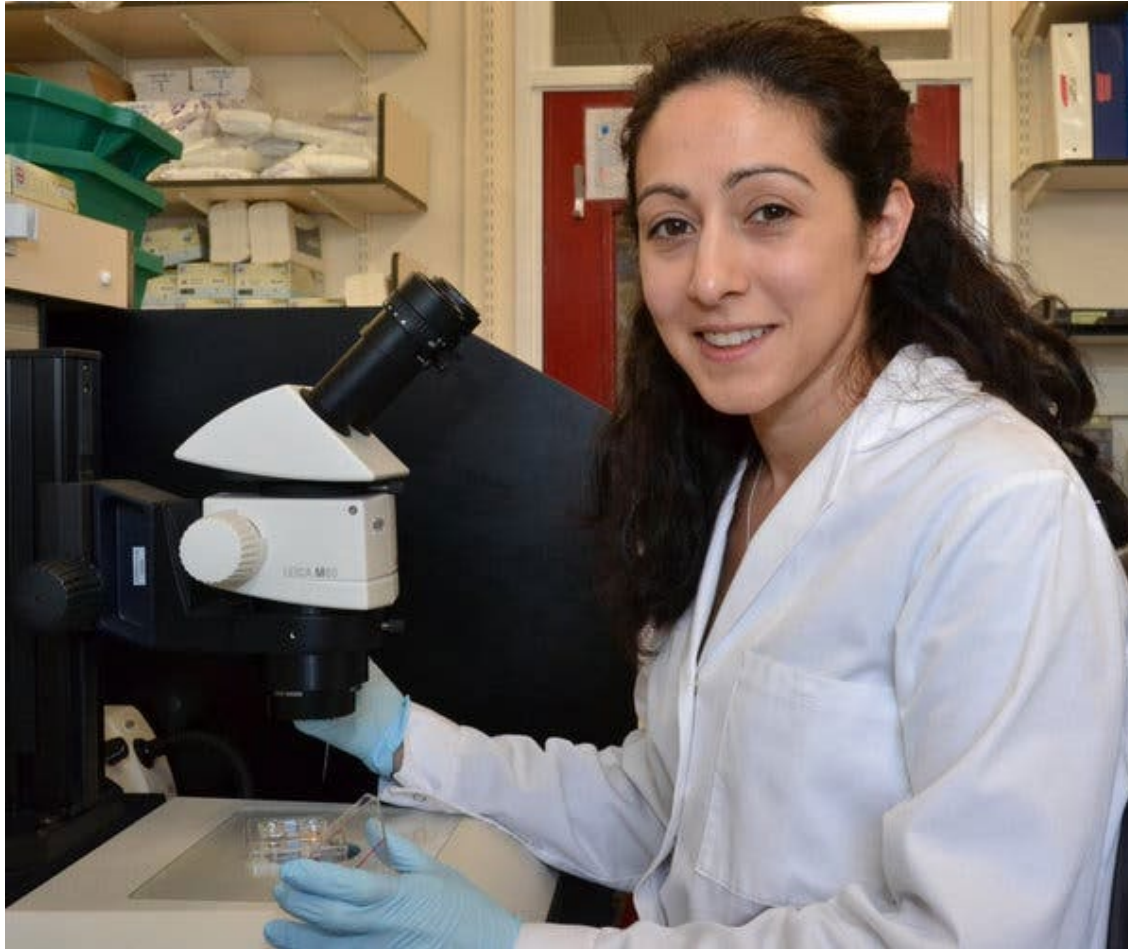


When is CRISPR introduced?

There are several cells that make up an embryo.

So we need to make sure that every cell has the editing tool.

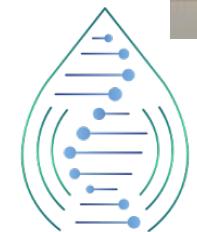




Dr Kathy Niakan, Cambridge University.

First in the UK to be granted permission to use CRISPR in embryos.

[Why scientists are editing genes in human embryos \(youtube.com\)](https://www.youtube.com/watch?v=...)



Most current research

A research fellow at Oxford University designed a project to introduce CRISPR into human embryos at two different stages of development.

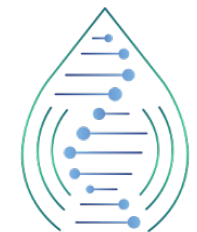


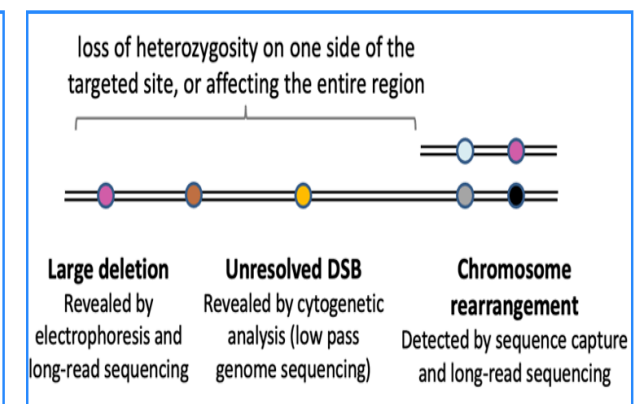
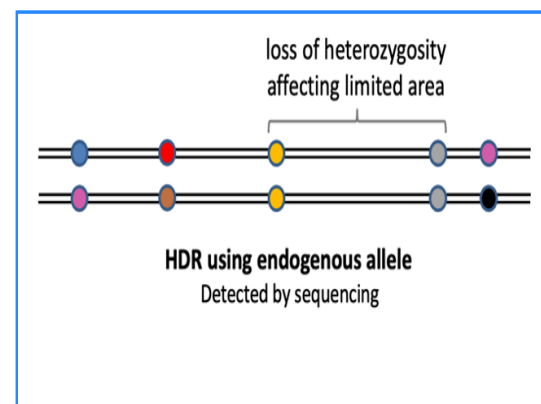
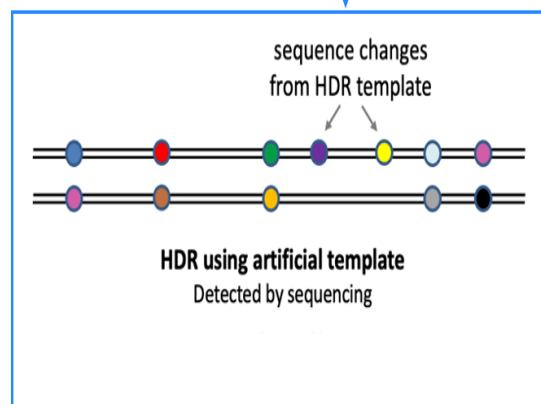
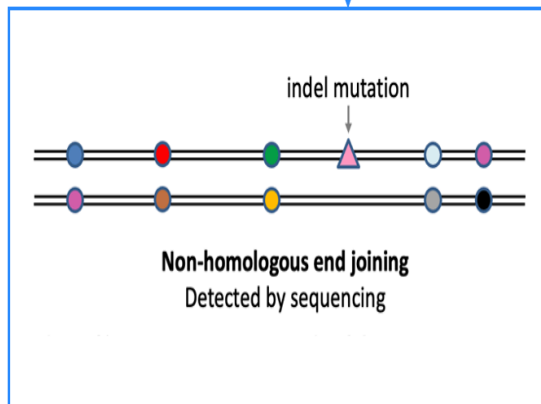
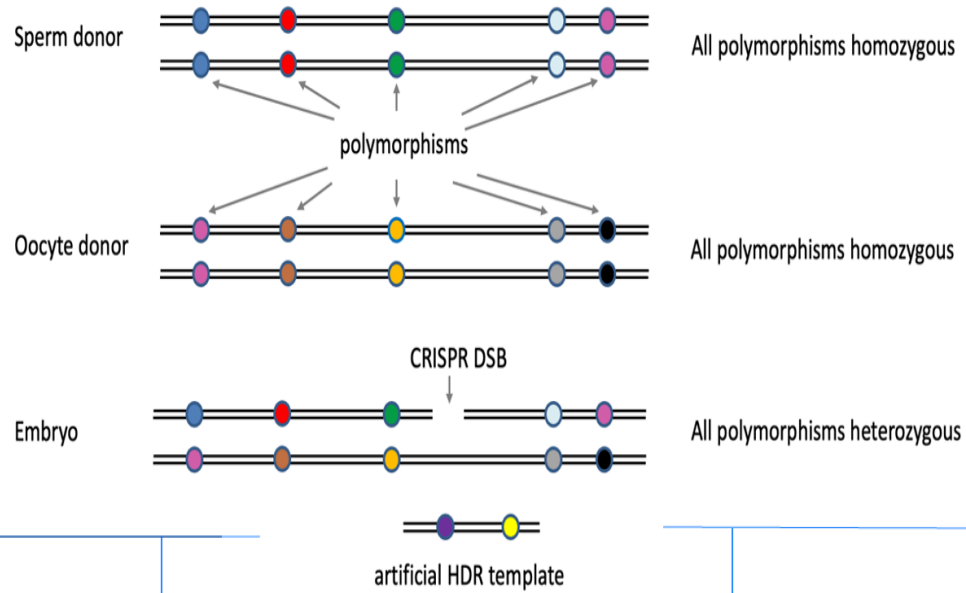
At fertilisation



Day 3 embryo

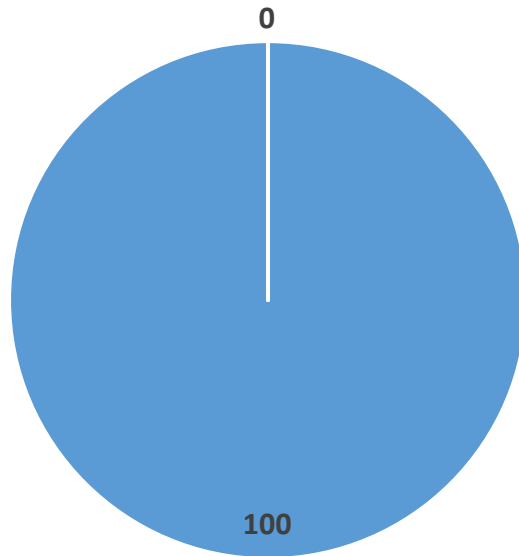
Unpublished data





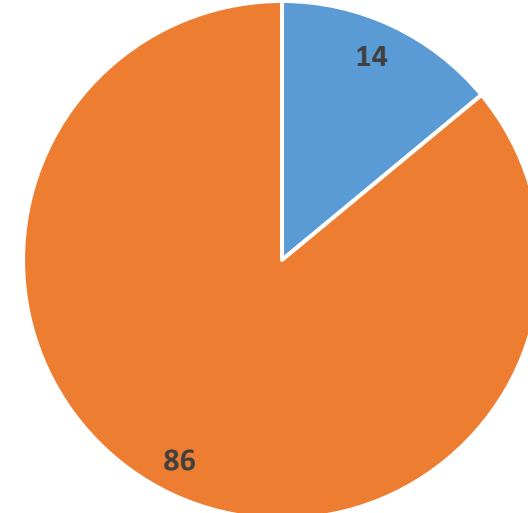
Results

Editing efficiency of ICSI group

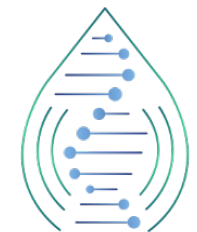


■ Successful ■ unsuccessful

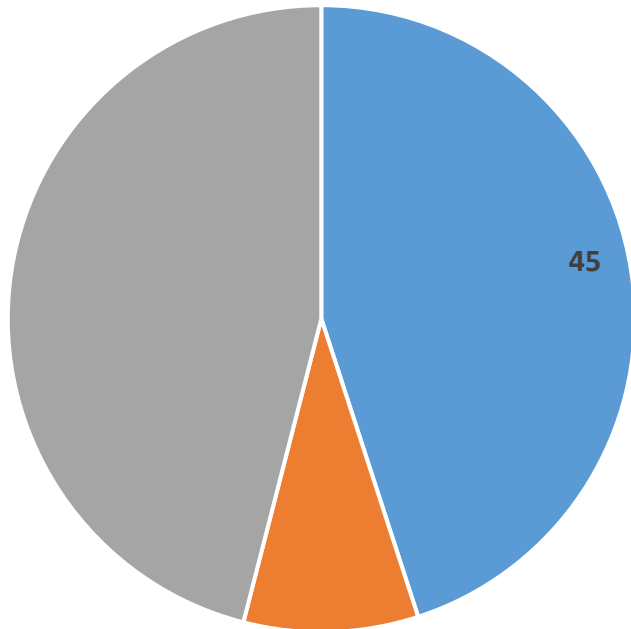
Editing efficiency of cleavage group



■ Successful ■ unsuccessful

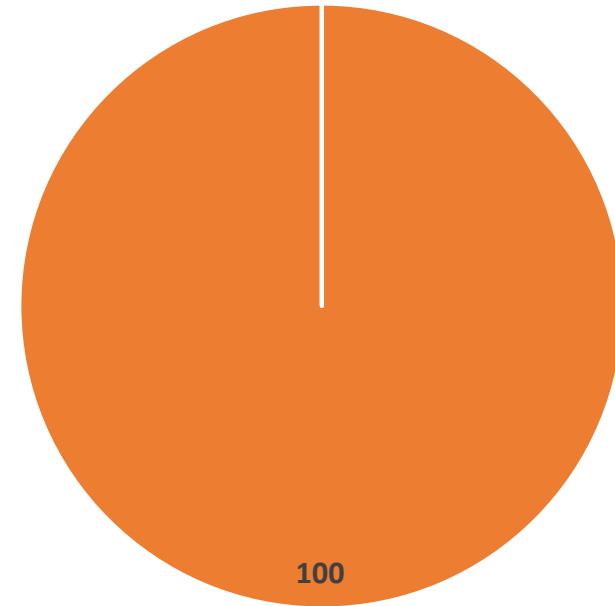


Repair mechanism of ICIS repair group

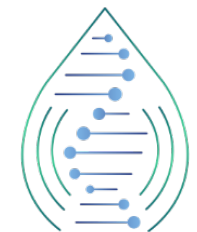


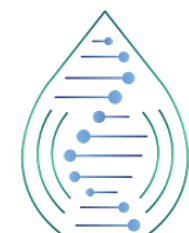
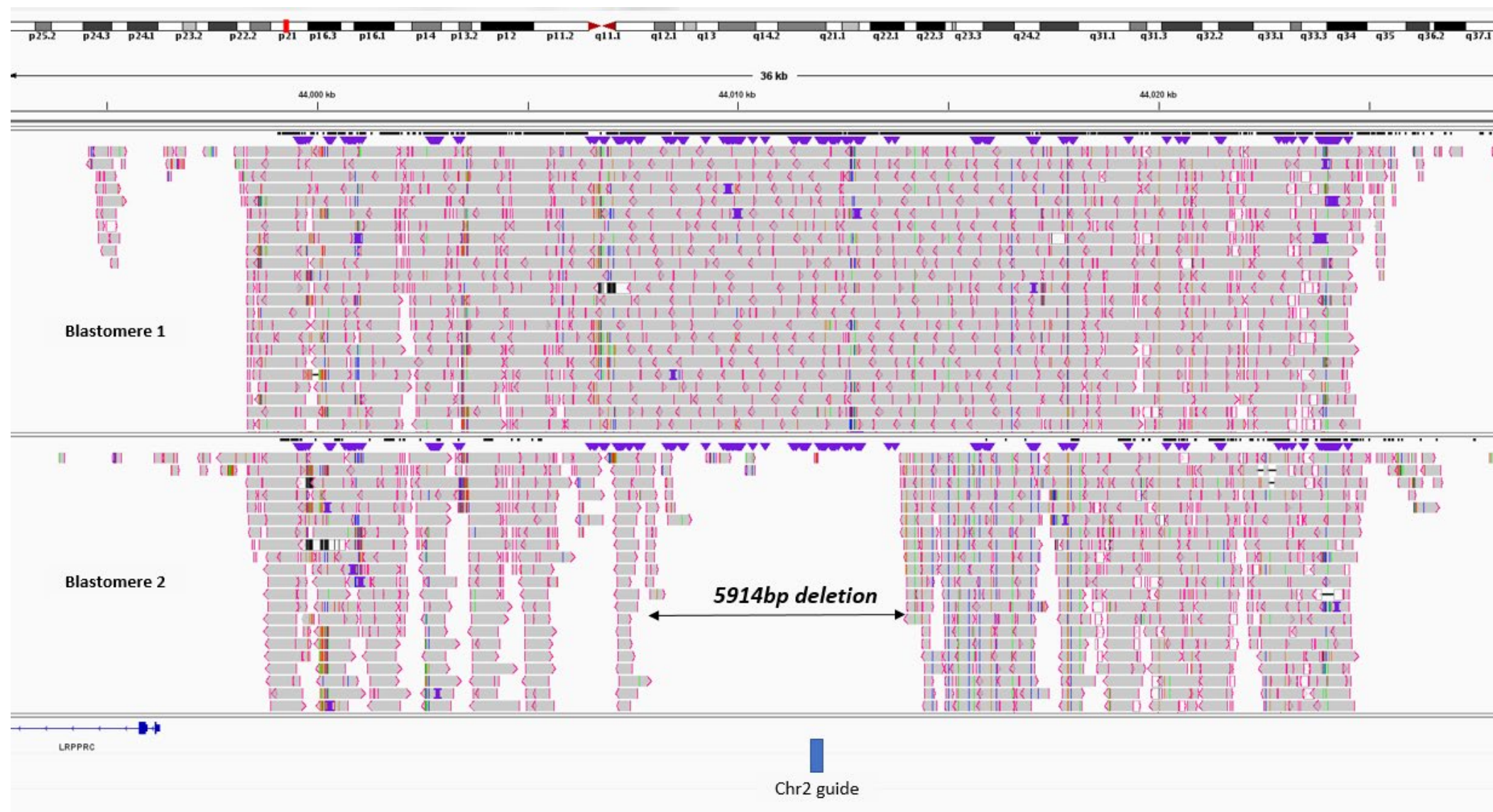
■ NHEJ ■ HDR ■ No repair

Repair mechanism of cleavage group



■ NHEJ ■ HDR ■ No repair





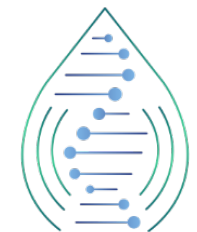
Conclusion

Successfully developed a method to detect large deletions.

Embryos at an early stage have little capacity for DNA repair.

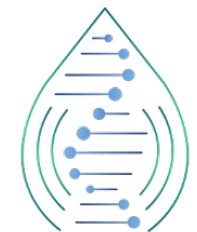
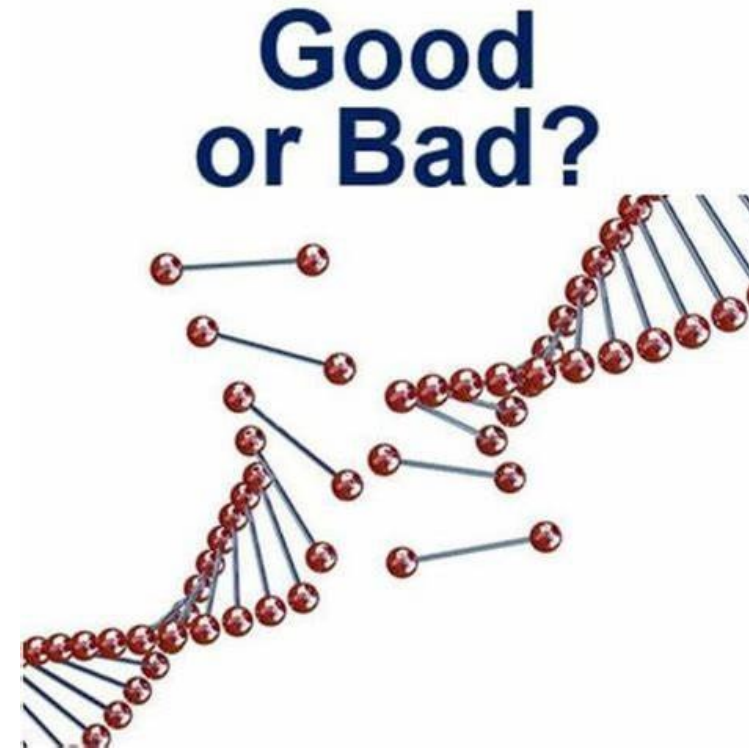
Embryonic genome activation is crucial.

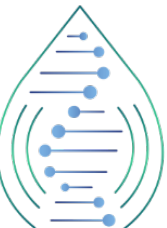
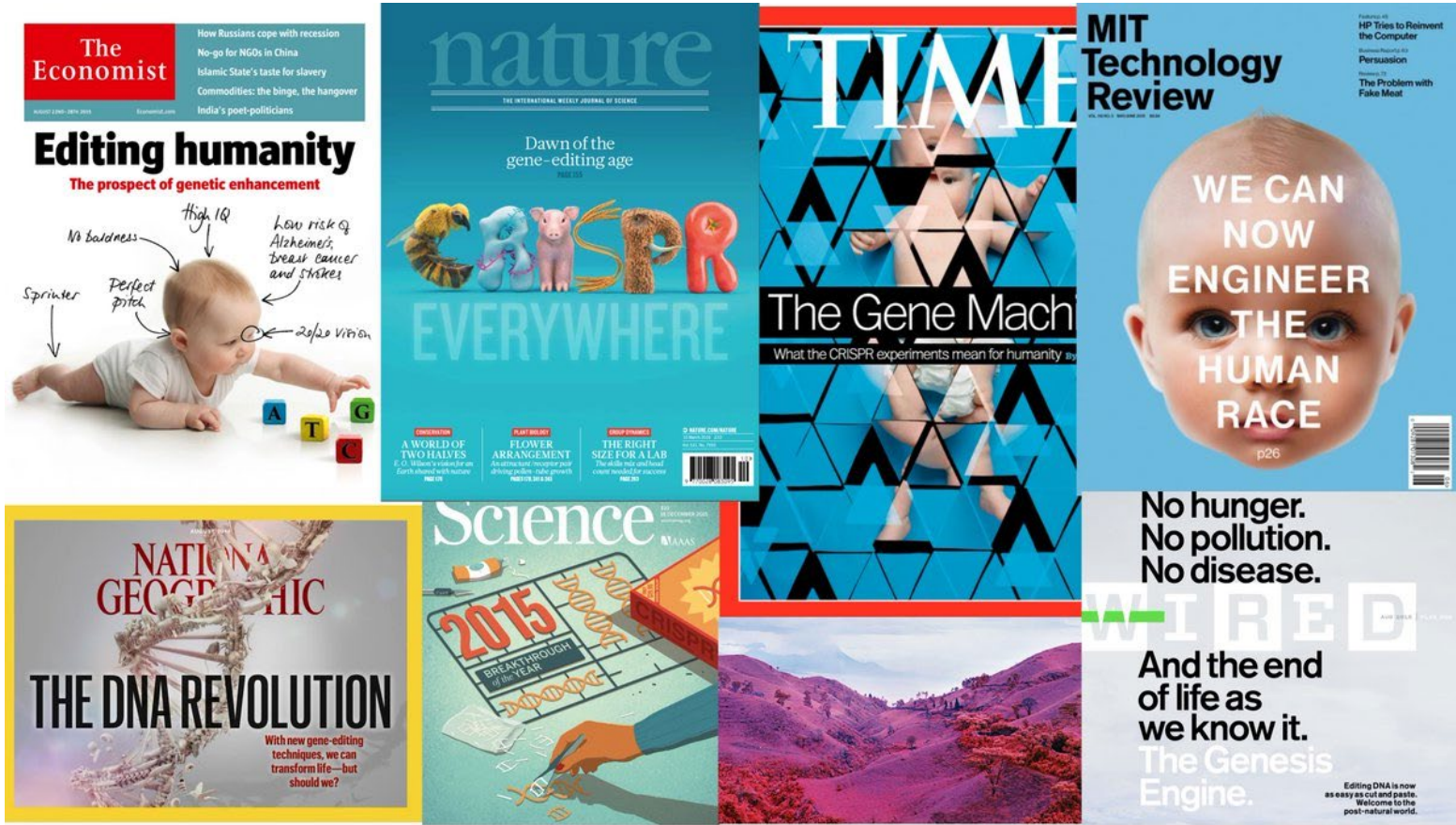
HDR most desired but NHEJ is the dominant repair mechanism.



Why are there so many restrictions to germline gene editing?

- Off-target effects
- Mosaicism
- Unknown long-term effects
- Ethical concerns
- Protecting public trust



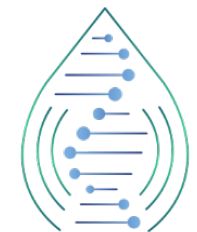


Controversial event

In 2018 a Chinese scientist – He Jiankui – created the first CRISPR babies.

He aimed to disable the CCR5 gene.

Twin girls were born known as Lulu and Nana as well as a third child, Amy.



He Jiankui's Research on CRISPR Babies Dubbed 'Delusional and Outrageous' as Experts See Manuscripts for First Time

Published Dec 03, 2019 at 9:23 AM EST

Birth of Twins After Genome Editing for HIV Resistance

Jinzhou Qin^{1,2,#}, Yangran Chen^{1,#}, Xiaoqing Zhou^{1,#}, Shuo Song^{1,#}, Kaijing Chen¹, Rui Chen¹, Yuanlin Chen¹, Hua Bai³, Michael W. Deem⁴, Jiankui He^{1,*}

¹ Department of Biology, Southern University of Science and Technology, Shenzhen, 518055, China.

² Department of Human Reproductive Medicine Center, Third Affiliated Hospital of Shenzhen University, Shenzhen 518001, China.

³ BaiHuaLin China People Living With HIV/AIDS Alliance, Cuipingli 10-132, Tongzhou, Beijing, China

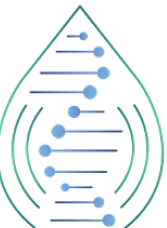
⁴ Departments of Bioengineering and Physics & Astronomy, Center for Theoretical Biological Physics, and Graduate Program in Systems, Synthetic, and Physical Biology, Rice University, Houston, TX 77005, USA.

*Corresponding Author. Email: hejk@sustc.edu.cn

#Equal contribution

In emails sent to American [Nobel-winning scientist Craig Mello](#) in April 2018, He confirmed the initial pregnancy. In one email, obtained by the Associated Press, He wrote: "Good news! The woman is pregnant, the genome editing a success! The embryo with CCR5 gene edited was transplanted to the woman 12 days ago, and today the pregnancy is confirmed."

Mello responded saying he did not want any involvement in the research saying he would "rather not be kept in the loop." He added there is no medical need for the experiments and that he is placing the health of the babies at risk. "I just don't see why you are doing this," Mello told He.



What is the future of embryo gene editing?

